ECON_OX 112 · PRINCIPLES OF MACROECONOMICS

Money demand, interest rates, and aggregate demand

November 3, 2026

Guilherme Paiva Pinto

Two FOMC Decisions

Earlier decision cards	June 15, 2022 · Case C	September 18, 2024 · Case E
Change in target range	Increase of 0.75 percentage point	Decrease of 0.50 percentage point
New target range	1.50–1.75%	4.75–5.00%

- The 2024 cut left the target range above its level after the 2022 increase.

FOMC, June 15, 2022 · FOMC, September 18, 2024

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- The 2024 cut left the target range above its level after the 2022 increase.
- How could either decision change the money people hold and the purchases firms make?

FOMC, June 15, 2022 · FOMC, September 18, 2024

Learning Objectives

- A · Understand and explain demand for money.
- B · Explain money-market equilibrium and the connection between money and interest rates.
- C · Interpret aggregate-demand shifts and demonstrate the effects of monetary policy.

15.1 Demand for Money

- Understand money demand as a desired stock for payments and liquidity.
- Explain how an investment’s return changes the opportunity cost of holding money.
- Distinguish a movement along money demand from a shift caused by other conditions.

Money Allocation Activity

- Use the separate PollEv activity: P1–P4. All amounts, rates, and contracts are hypothetical.
- Choose for the fictional household. Keep enough money for its upcoming bills.
- Keep your reasoning: we will use the first responses to build a money-demand comparison.

The Opportunity Cost of Holding Money

Holding $600 for one year	Ending balance	Interest forgone by holding money
Cash/checking at 0%	$600	—
One-year bond at 1%	$606	$6
One-year bond at 6%	$636	$36

- Rate of return = (ending value − initial value) ÷ initial value.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

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- A higher return on the alternative makes a liquid balance more costly to hold.

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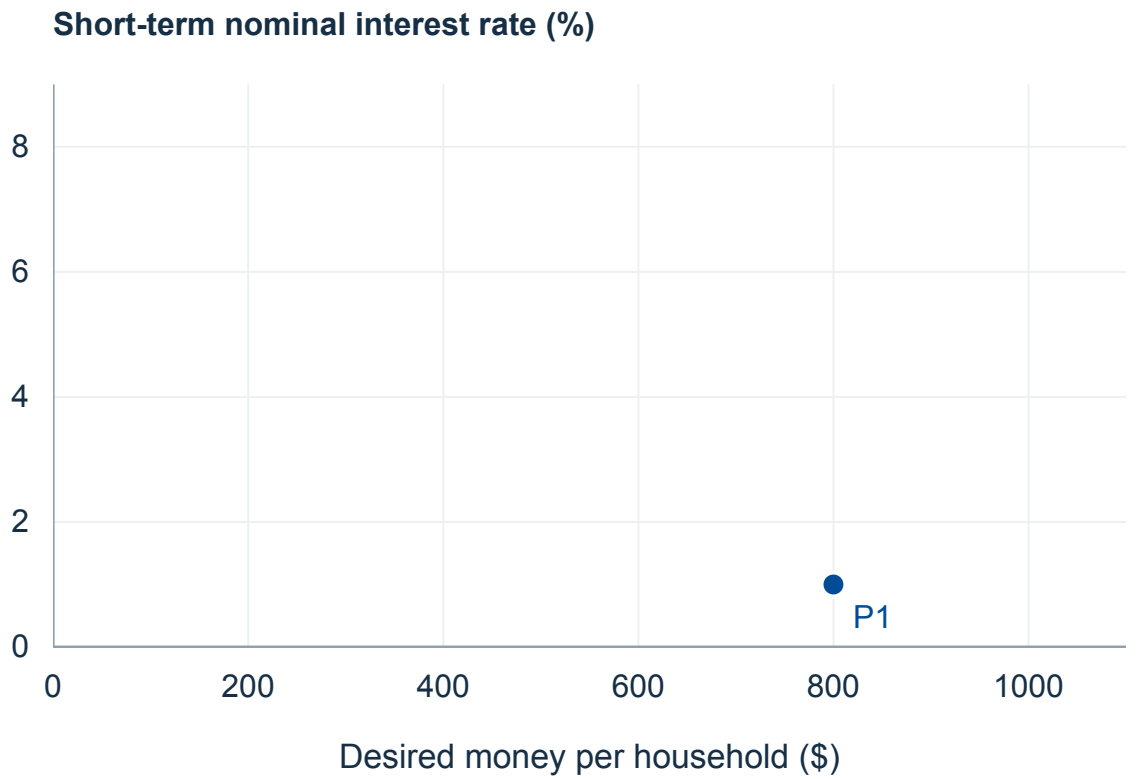
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- Rate of return = (ending value − initial value) ÷ initial value.
- A higher return on the alternative makes a liquid balance more costly to hold.
- Compare with the return paid on money itself; today the activity holds that return at 0%.

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From Choices to Money Demand



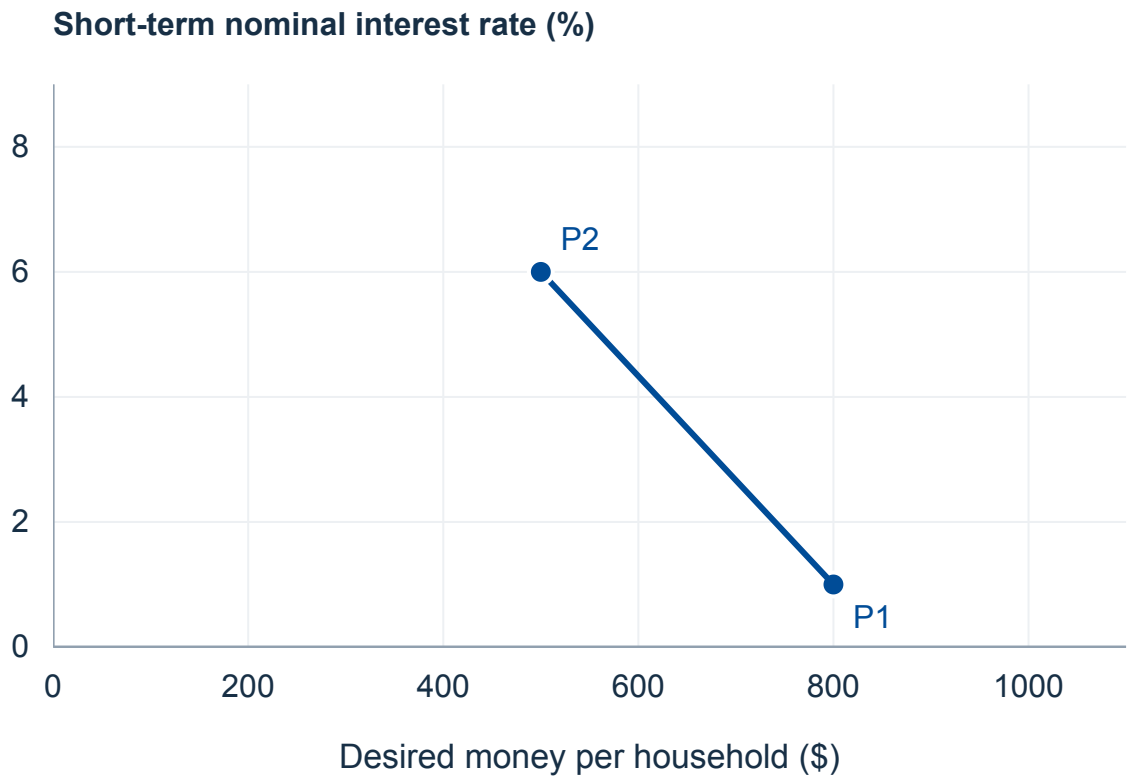
Schematic comparison; enter class counts to plot observed means.

P1–P2 compare returns; P3 changes bills at the same rate.

Same household, different return

P1 → P2: only the one-year return rises, from 1% to 6%.

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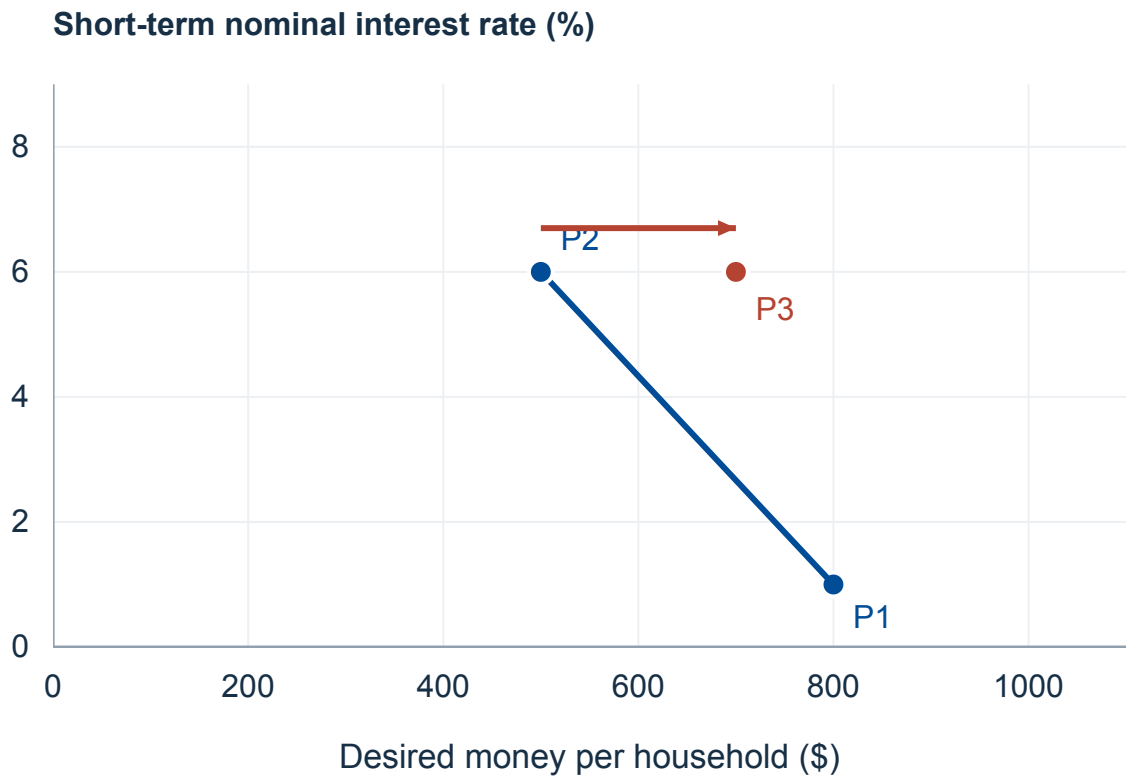
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Compare the two choices

At a higher short-term rate, the model predicts a smaller quantity of money demanded.

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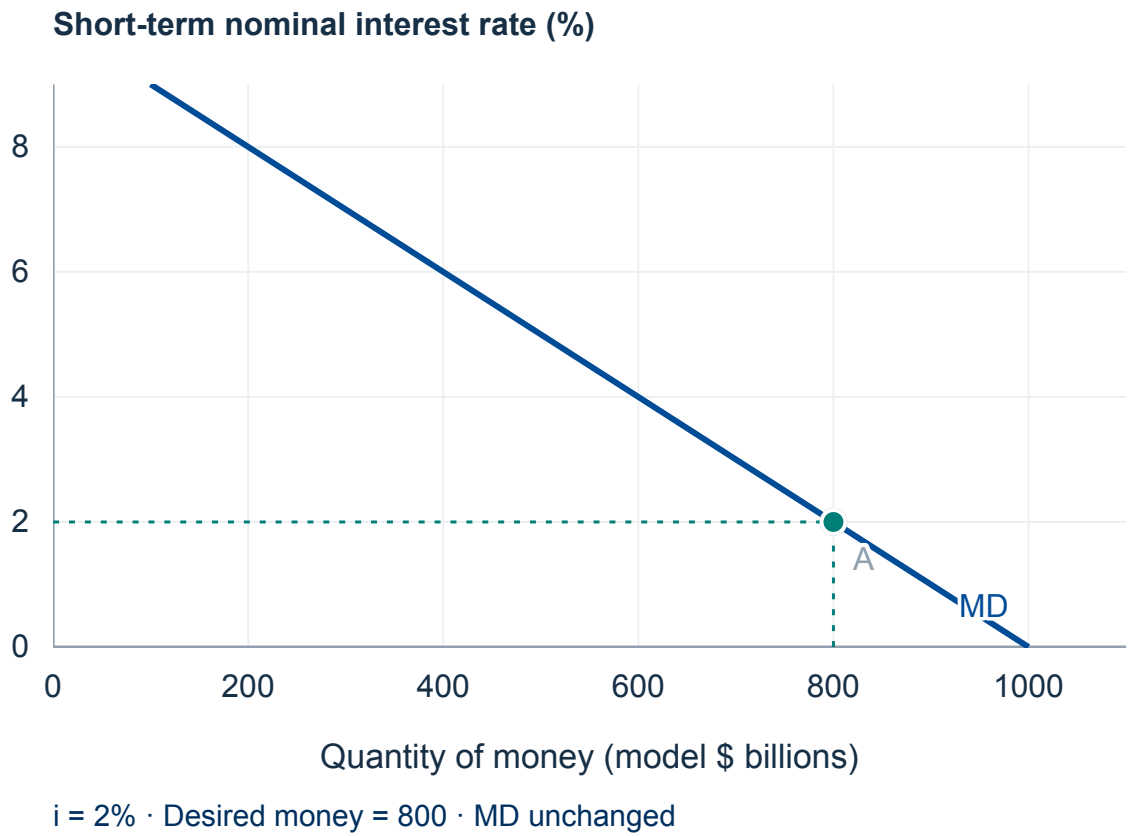
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At a higher short-term rate, the model predicts a smaller quantity of money demanded.

A different cause: higher bills

P2 → P3: bills become more expensive at the same 6% rate. That is a demand shift.

The Demand for Money



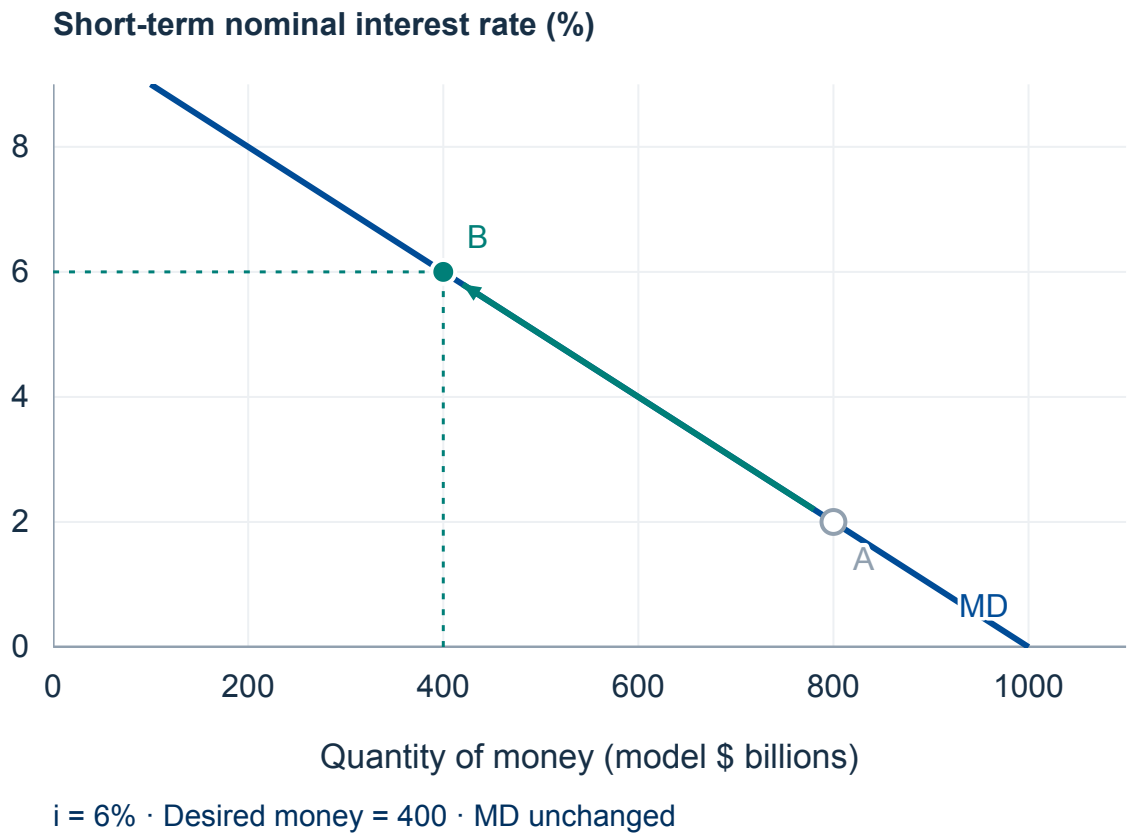
i = 2% · Desired money = 800 · MD unchanged

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

Hold the other determinants fixed

Hold the price level, real GDP, payment arrangements, and money's own return fixed.

The Demand for Money



i = 6% · Desired money = 400 · MD unchanged

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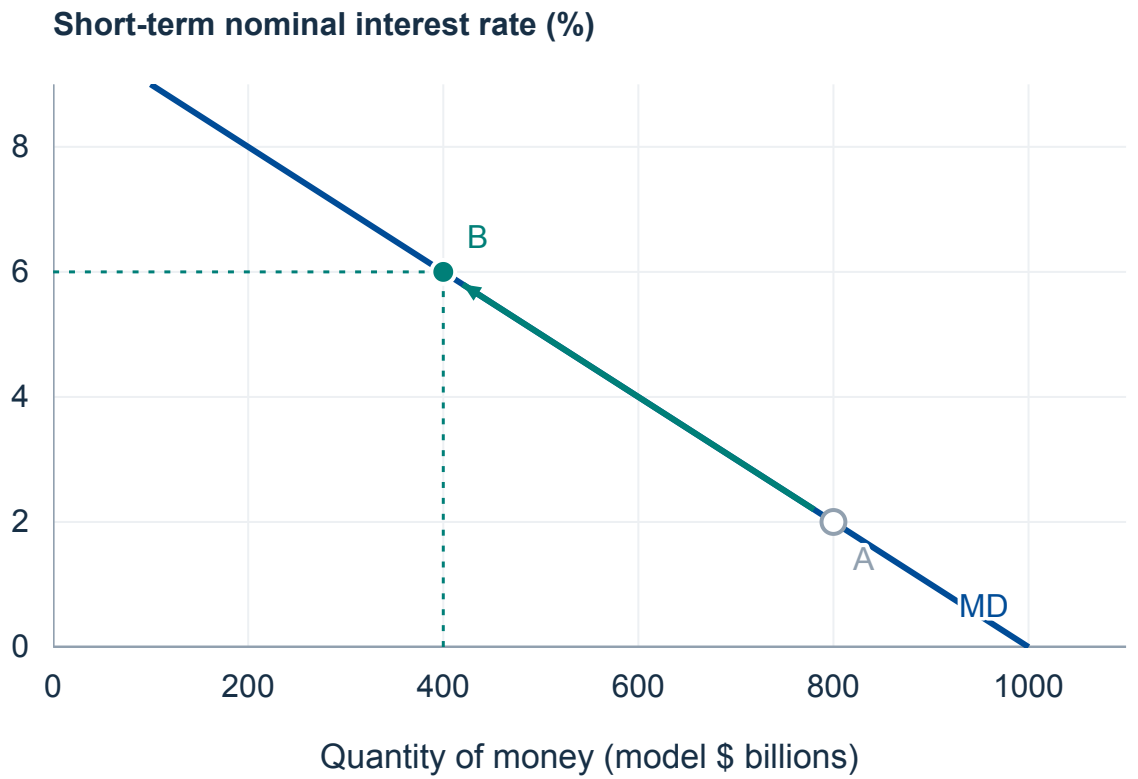
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The opportunity cost rises

A rise in the short-term nominal interest rate raises the cost of holding money.

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Read the new desired balance

Move upward along MD: less money is desired for liquidity at the higher rate.

Shifts in Money Demand

- Price level ↑ → the same real purchases require more dollars → MD shifts right.

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- Easier access to short-term credit can reduce the balances needed between paydays → MD may shift left.
- Faster, reliable banking and payments can reduce idle balances → MD may shift left.
- Institutions that require larger cash buffers can raise desired balances → MD shifts right.

Money Demand Checkpoint

1. The short-term interest rate rises while prices, real GDP and payment arrangements stay fixed. Movement along MD or a shift?

2. At the same interest rate, the same basket of bills costs more dollars. Movement or shift?

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2. At the same interest rate, the same basket of bills costs more dollars. Movement or shift?

> MD shifts right: people need larger nominal balances for the same real purchases.

15.2 Money and Interest Rates

- Explain the federal funds rate and the Fed’s influence on short-term rates.
- Interpret money-market equilibrium and shifts in the supply of money.
- Explain why longer-term rates reflect expectations and need not move one-for-one with today’s policy rate.

The Federal Funds Rate

- Some banks lend reserve balances to other institutions overnight. The federal funds rate is the rate on those loans.

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

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Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

The Federal Funds Rate

- Some banks lend reserve balances to other institutions overnight. The federal funds rate is the rate on those loans.
- The FOMC announces a target range; the market determines the effective rate.
- The Fed influences this overnight rate, which helps shape other short-term rates and broader financial conditions.

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

From a Fed Purchase to Money Supply

1 Recall the previous class: the Fed buys a Treasury security from a nonbank seller.

Today’s ample-reserves system relies mainly on administered rates such as IORB to steer overnight rates. A rate cut does not require a matching bond purchase.

Federal Reserve Board: implementation and IORB

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Federal Reserve Board: implementation and IORB

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3. The monetary base rises; the seller now holds a deposit instead of a security. In this model, MS shifts right.

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Federal Reserve Board: implementation and IORB

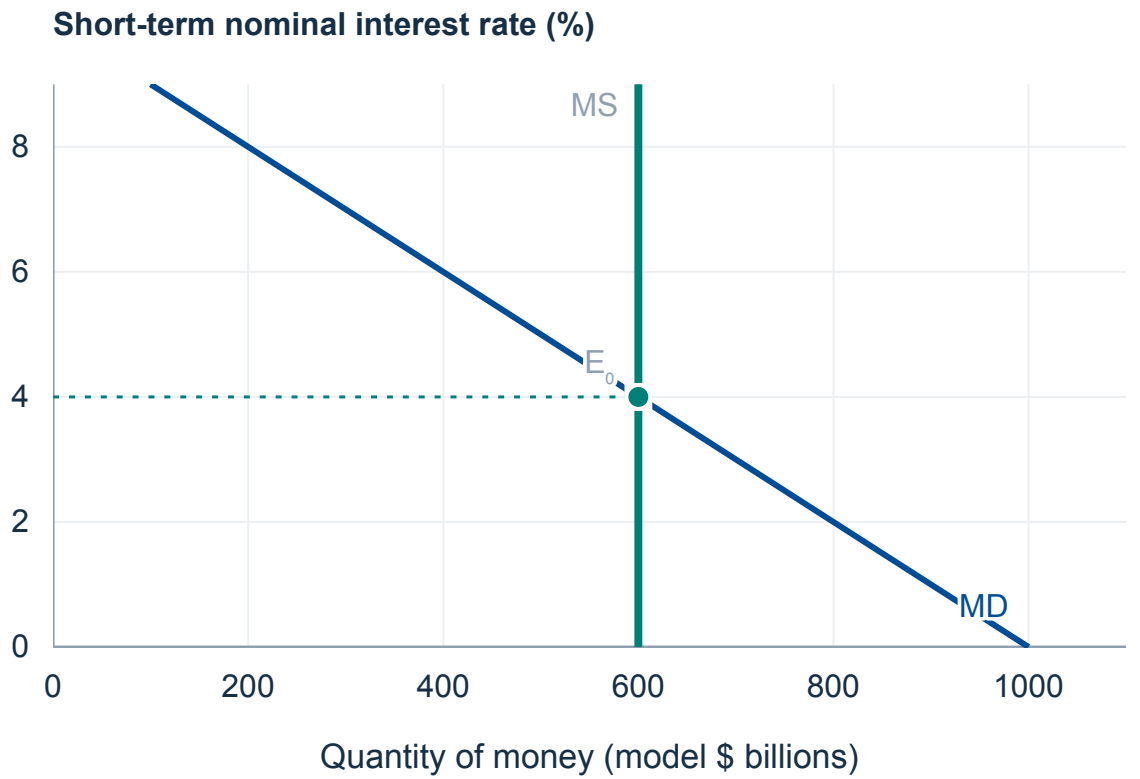
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2. The payment credits the seller’s bank with reserves; the bank credits the seller’s deposit.
3. The monetary base rises; the seller now holds a deposit instead of a security. In this model, MS shifts right.
4. The interest rate adjusts along the existing MD curve. Policy did not directly shift MD.

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Federal Reserve Board: implementation and IORB

Expansionary Monetary Policy



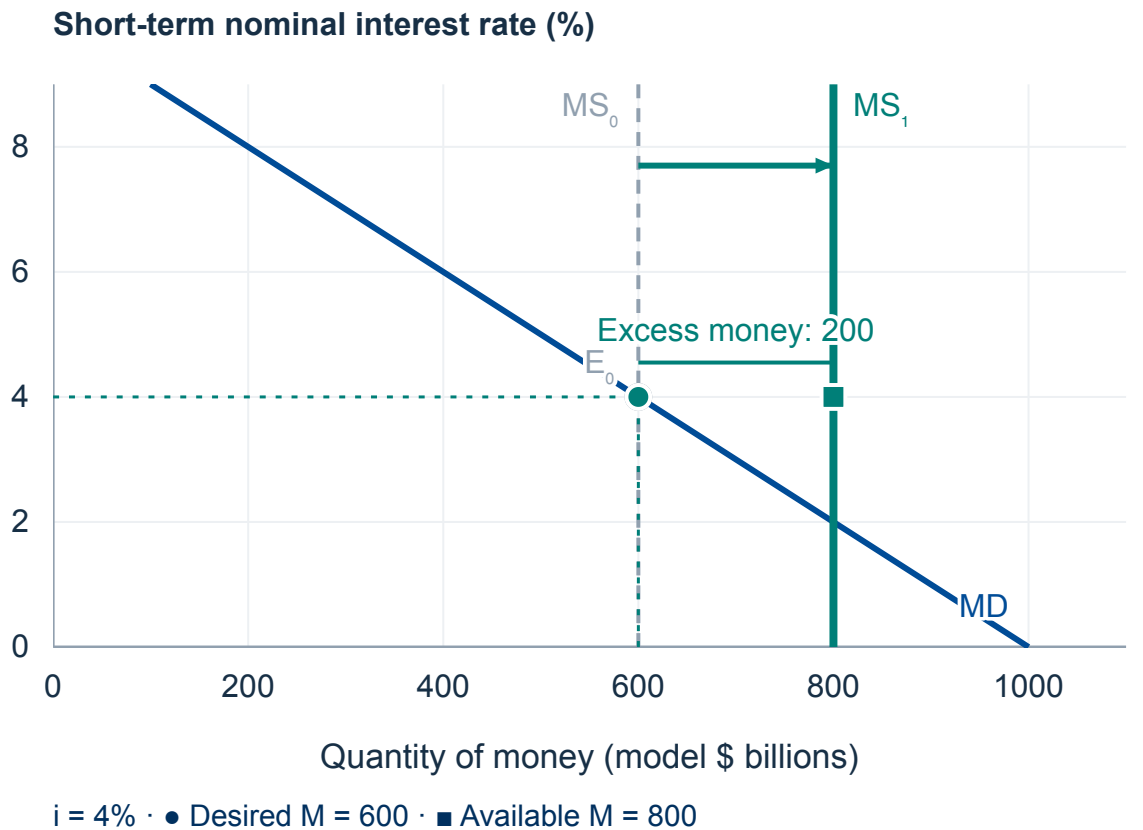
i = 4% · ● Desired M = 600 · ■ Available M = 600

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

Initial equilibrium

Start at $MS_0 = 600$ and $i_0 = 4\%$.

Expansionary Monetary Policy



i = 4% · ● Desired M = 600 · ■ Available M = 800

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

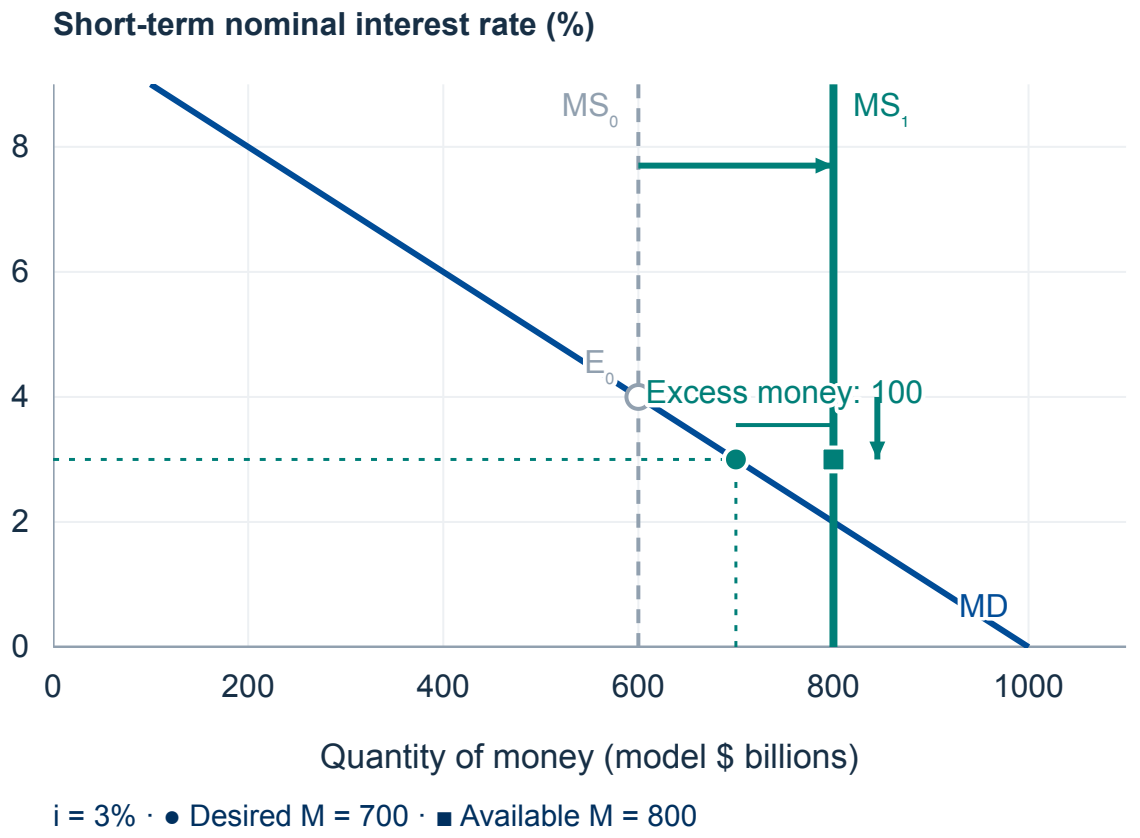
Initial equilibrium

Start at MS_0 = 600 and i_0 = 4%.

Money supply increases

In the money-supply model, a purchase raises MS to 800.

Expansionary Monetary Policy



i = 3% · ● Desired M = 700 · ■ Available M = 800

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

Initial equilibrium

Start at MS_0 = 600 and i_0 = 4%.

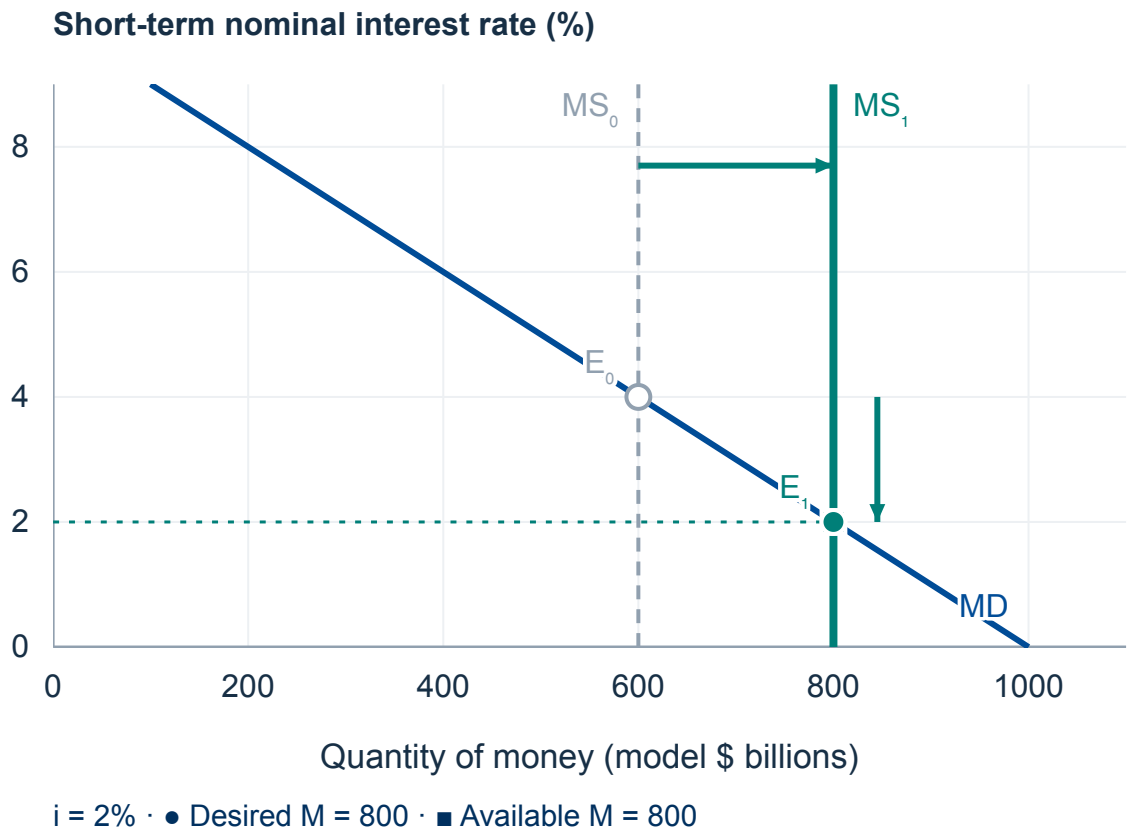
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Asset-market adjustment

At 4%, people hold more money than they desire; they seek bonds and other assets.

Expansionary Monetary Policy



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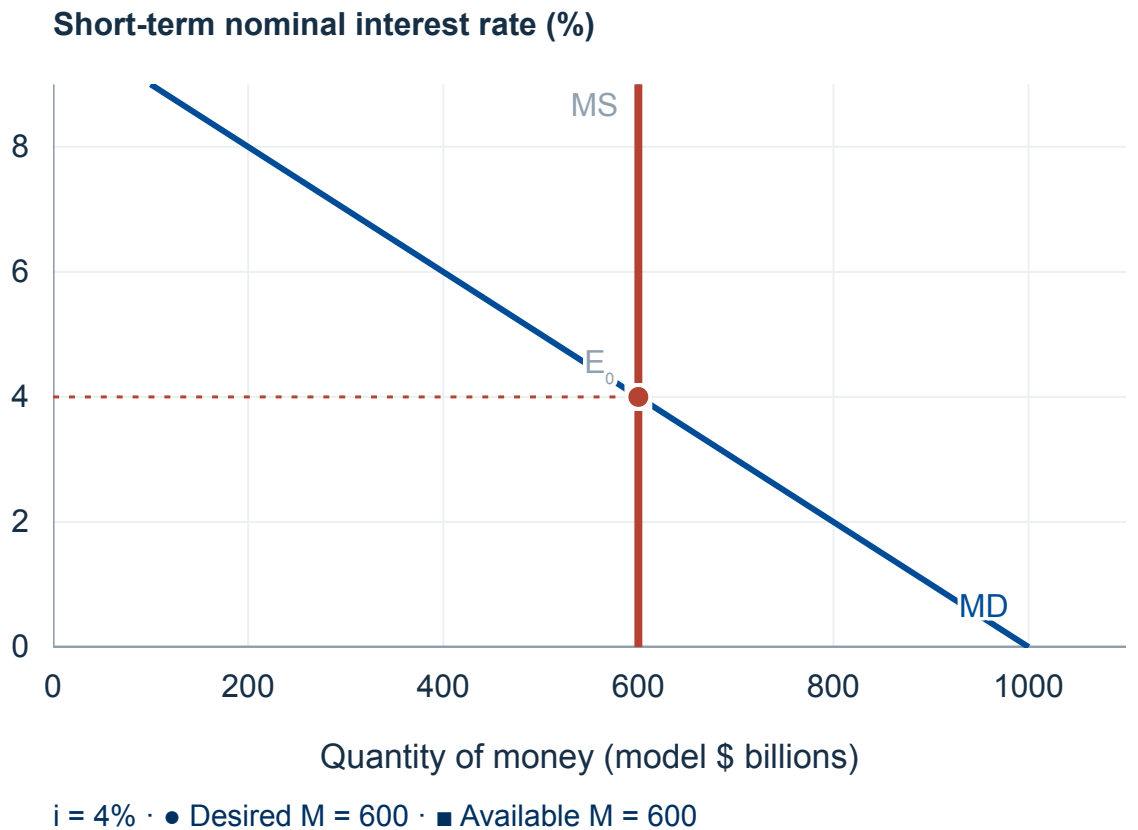
Asset-market adjustment

At 4%, people hold more money than they desire; they seek bonds and other assets.

New equilibrium

Bond prices rise and yields fall. New equilibrium: i_1 = 2%, M_1 = 800.

Contractionary Monetary Policy



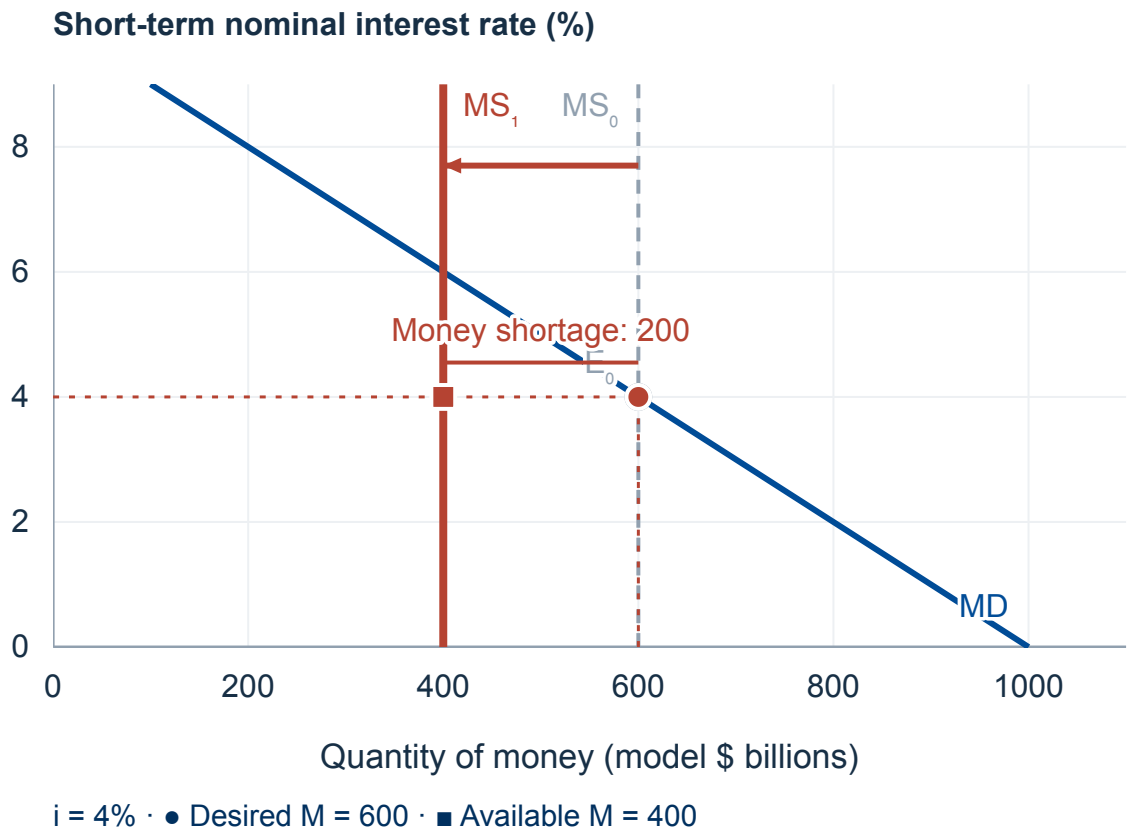
i = 4% · ● Desired M = 600 · ■ Available M = 600

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

Initial equilibrium

Start at $MS_0 = 600$ and $i_0 = 4\%$.

Contractionary Monetary Policy



i = 4% · ● Desired M = 600 · ■ Available M = 400

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

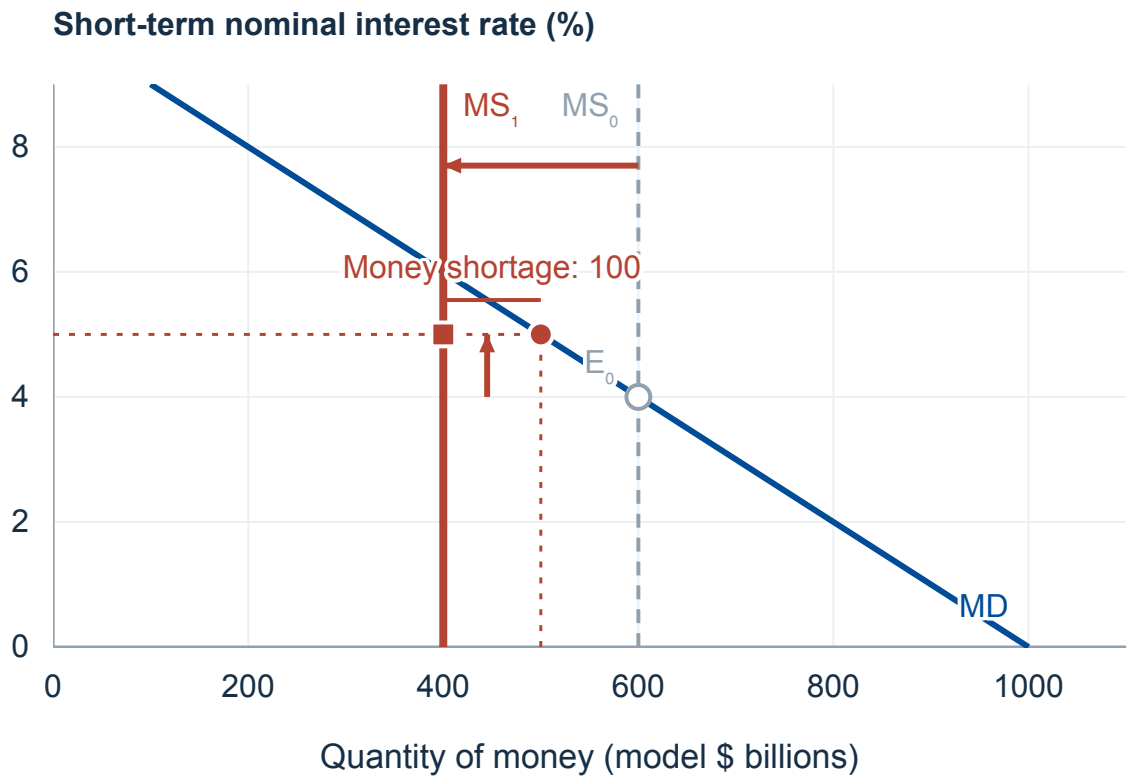
Initial equilibrium

Start at $MS_0 = 600$ and $i_0 = 4\%$.

Money supply decreases

In the money-supply model, a sale reduces MS to 400.

Contractionary Monetary Policy



i = 5% · ● Desired M = 500 · ■ Available M = 400

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

Initial equilibrium

Start at MS_0 = 600 and i_0 = 4%.

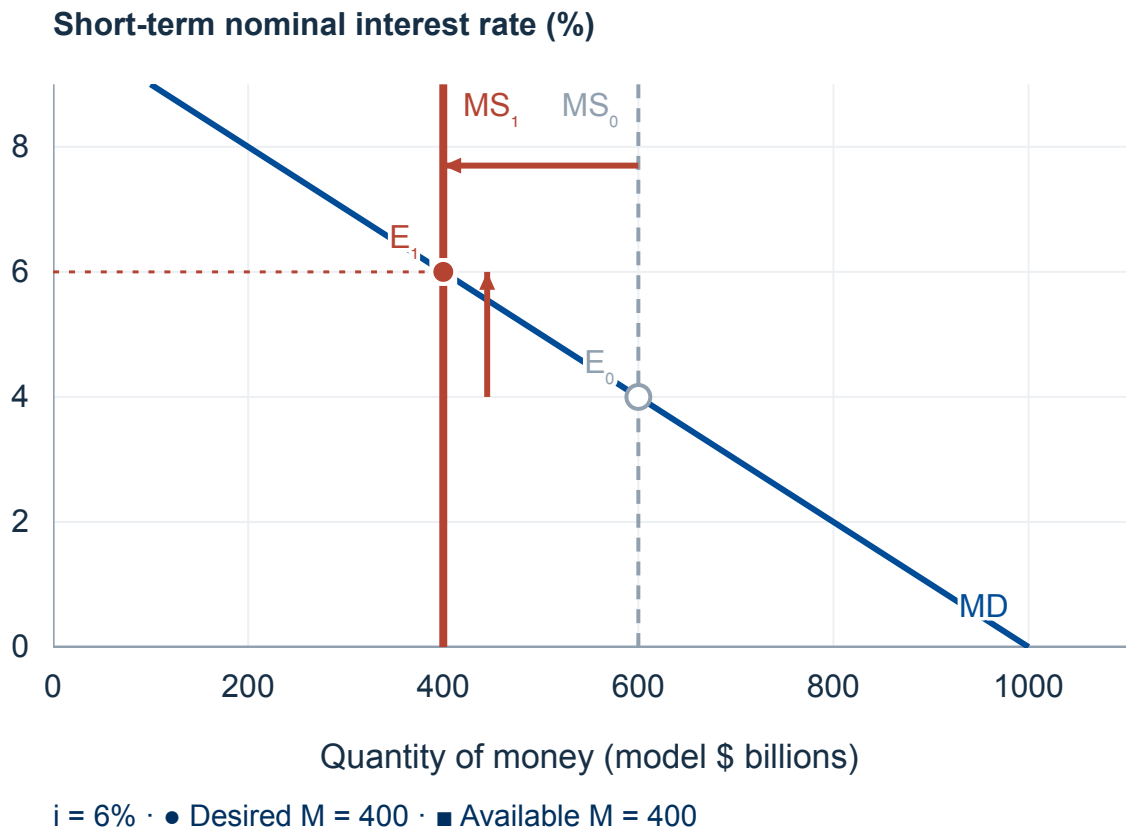
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Asset-market adjustment

At 4%, people desire more money than is available; they sell assets to obtain balances.

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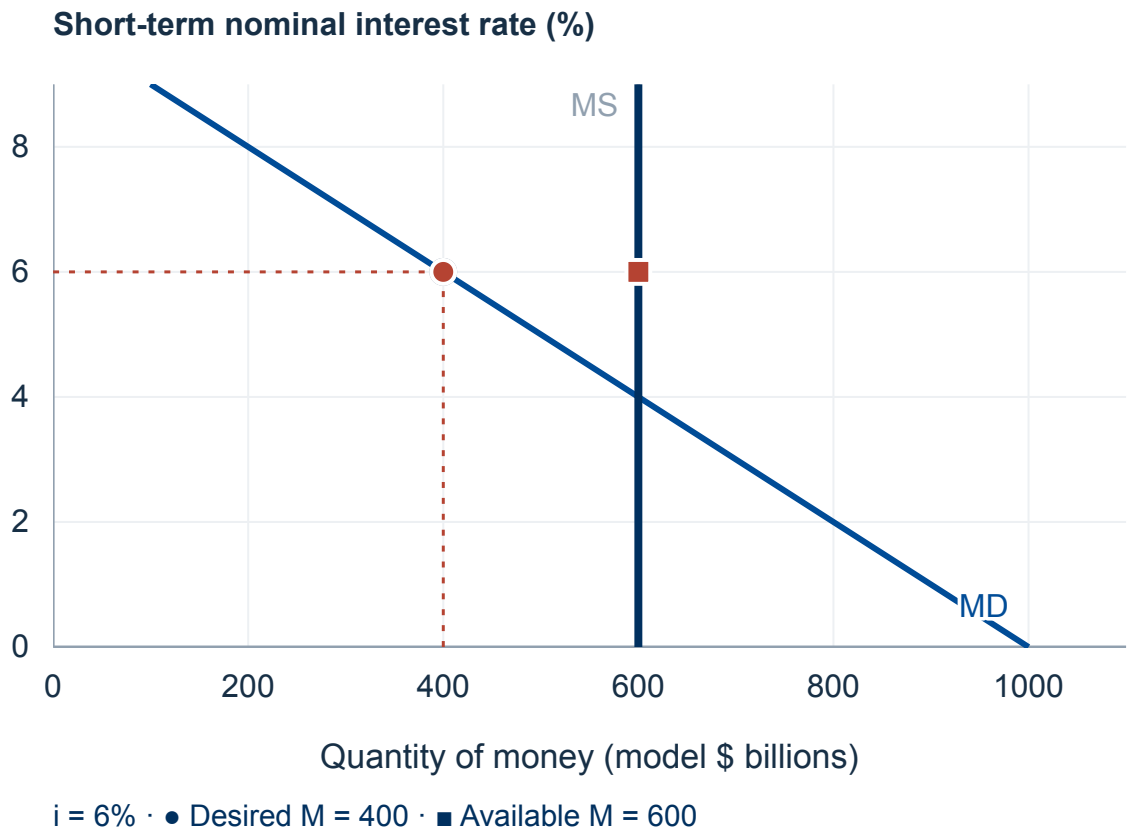
Asset-market adjustment

At 4%, people desire more money than is available; they sell assets to obtain balances.

New equilibrium

Bond prices fall and yields rise. New equilibrium: $i_1 = 6\%$, $M_1 = 400$.

Adjustment from a High Interest Rate



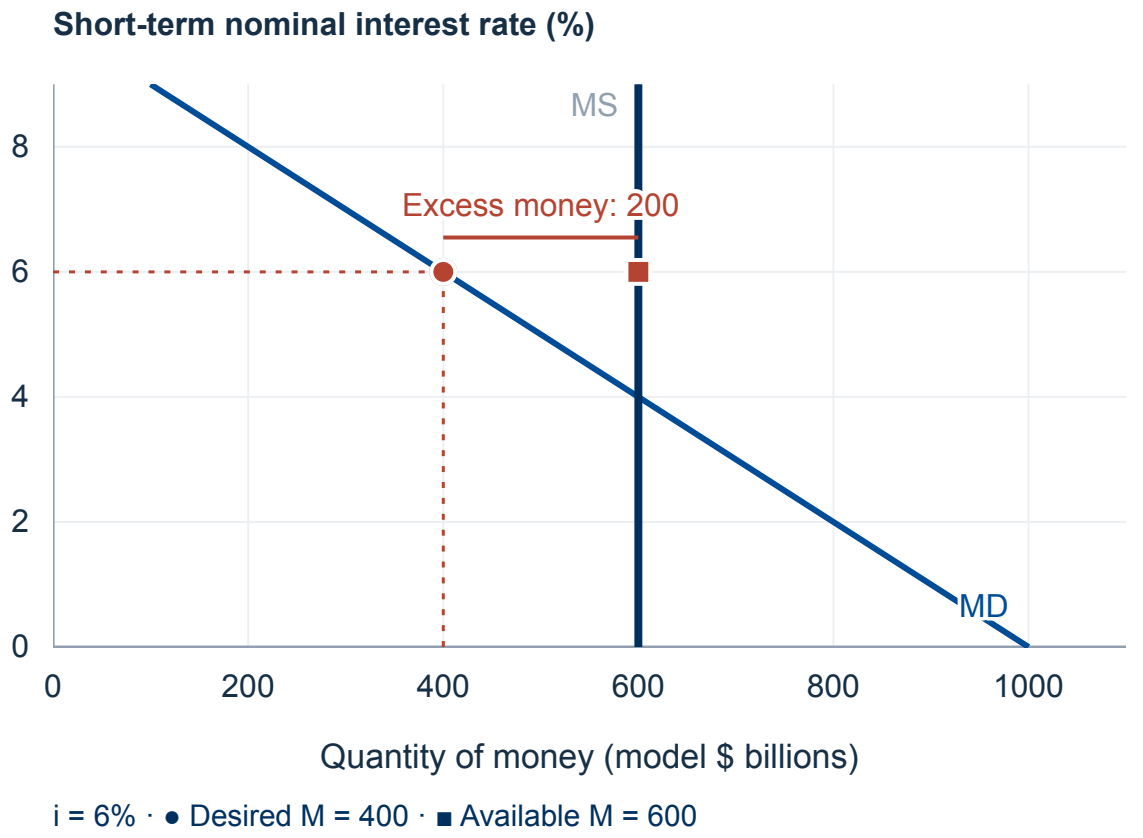
i = 6% · ● Desired M = 400 · ■ Available M = 600

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

A rate above equilibrium

Suppose i = 6%: MD = 400 but MS = 600. Desired money holdings are below the available stock.

Adjustment from a High Interest Rate



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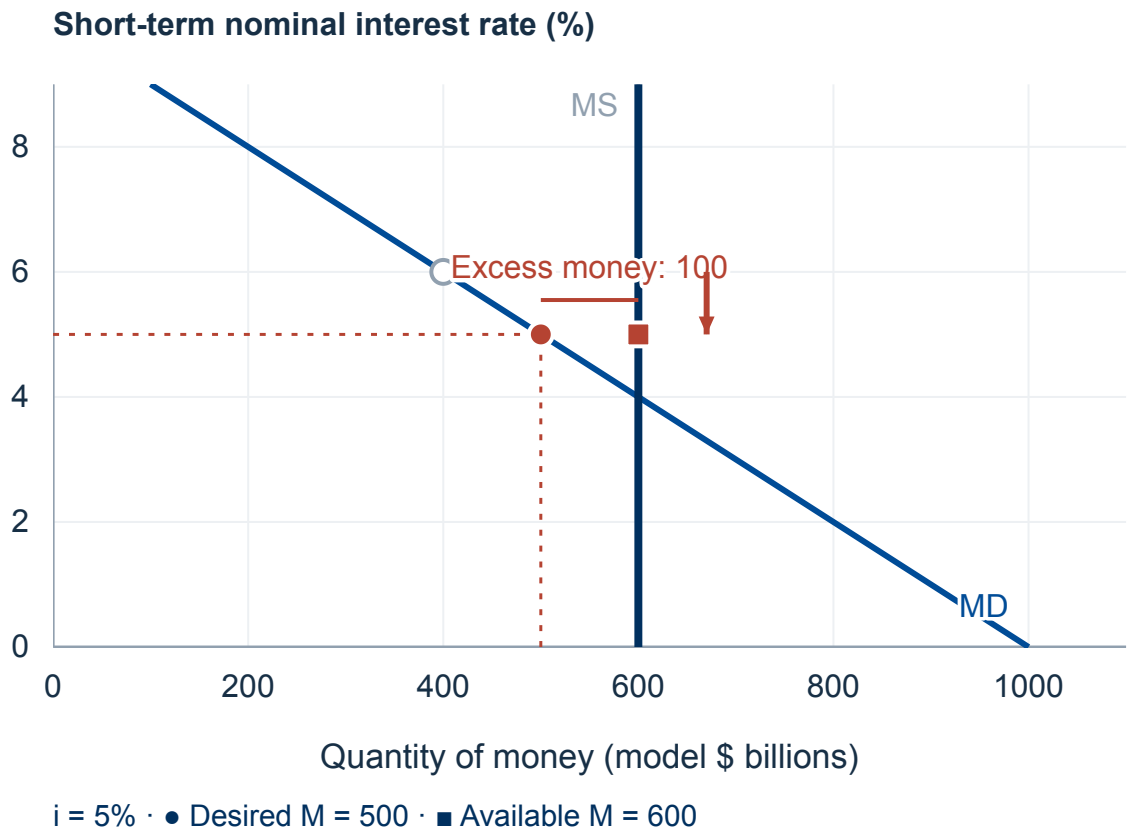
A rate above equilibrium

Suppose i = 6%: MD = 400 but MS = 600. Desired money holdings are below the available stock.

Excess money balances

People try to exchange excess balances for bonds. The money remains in the system when it changes hands.

Adjustment from a High Interest Rate



i = 5% · ● Desired M = 500 · ■ Available M = 600

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

A rate above equilibrium

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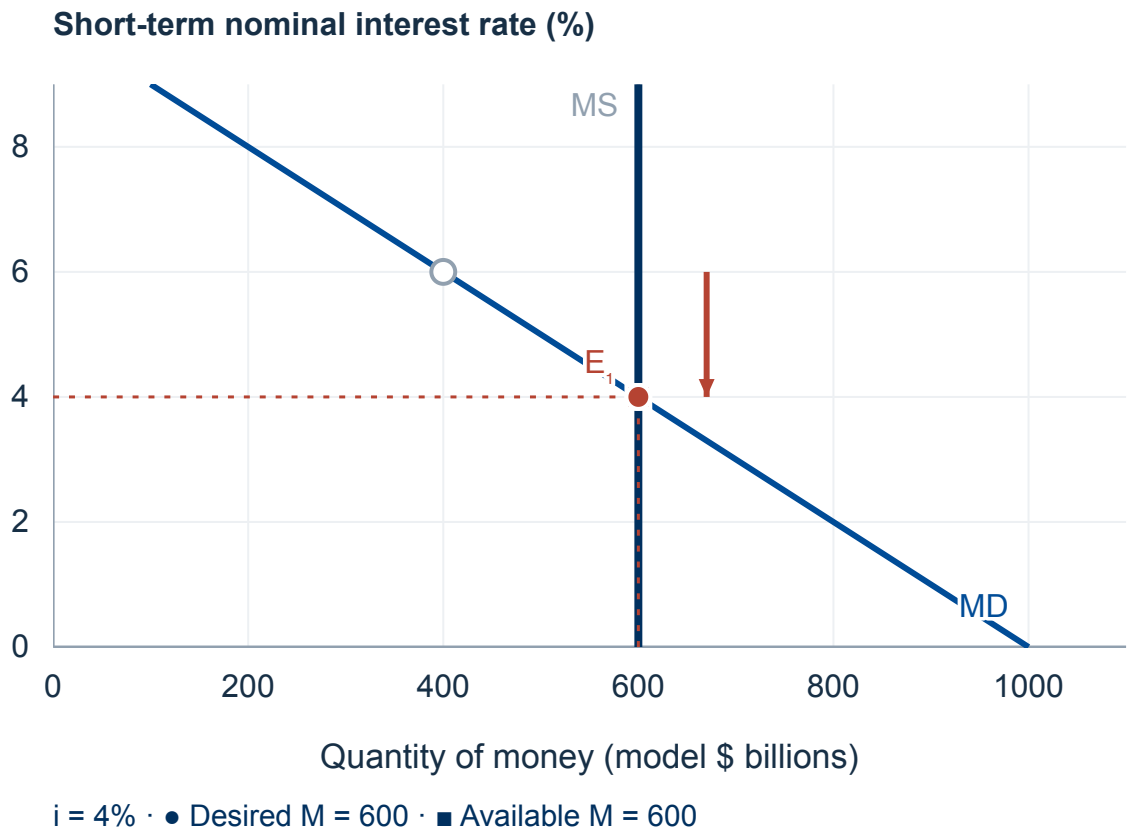
Excess money balances

People try to exchange excess balances for bonds. The money remains in the system when it changes hands.

Bond prices rise; yields fall

Bidding raises bond prices and lowers yields. A $100 one-year payoff rises from $94.34 to $96.15.

Adjustment from a High Interest Rate



i = 4% · ● Desired M = 600 · ■ Available M = 600

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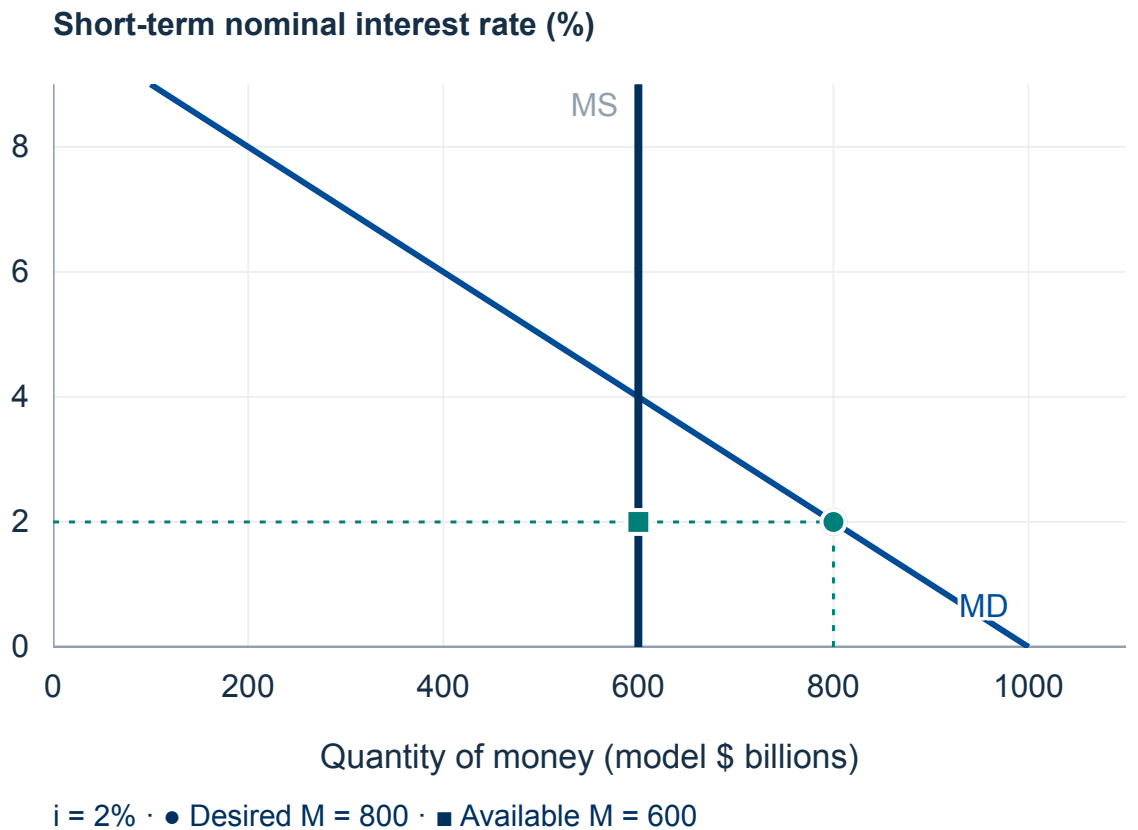
Bond prices rise; yields fall

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Money-market equilibrium

At i = 4%, desired and available money both equal 600: the adjustment pressure ends.

Adjustment from a Low Interest Rate



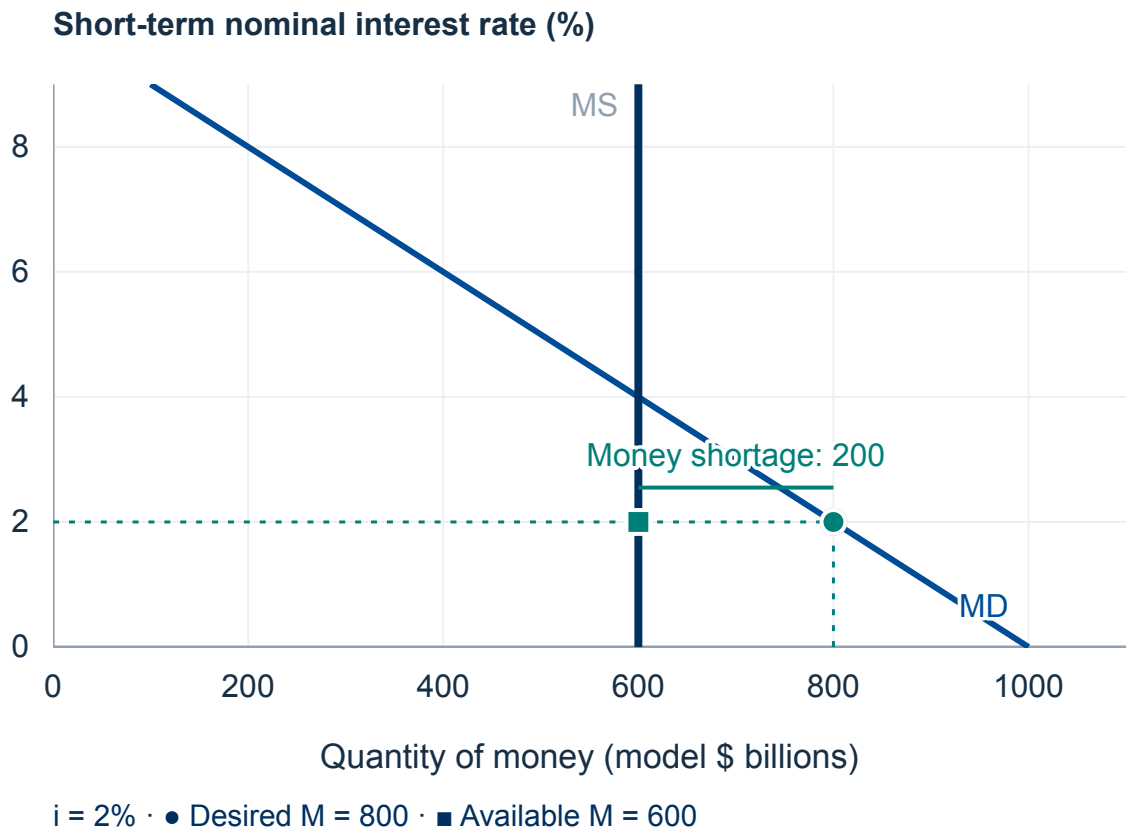
i = 2% · ● Desired M = 800 · ■ Available M = 600

MD = 1,000 − 100i. Fixed prices, real GDP and money's own return.

A rate below equilibrium

Suppose i = 2%: MD = 800 but MS = 600. People want more liquid balances.

Adjustment from a Low Interest Rate



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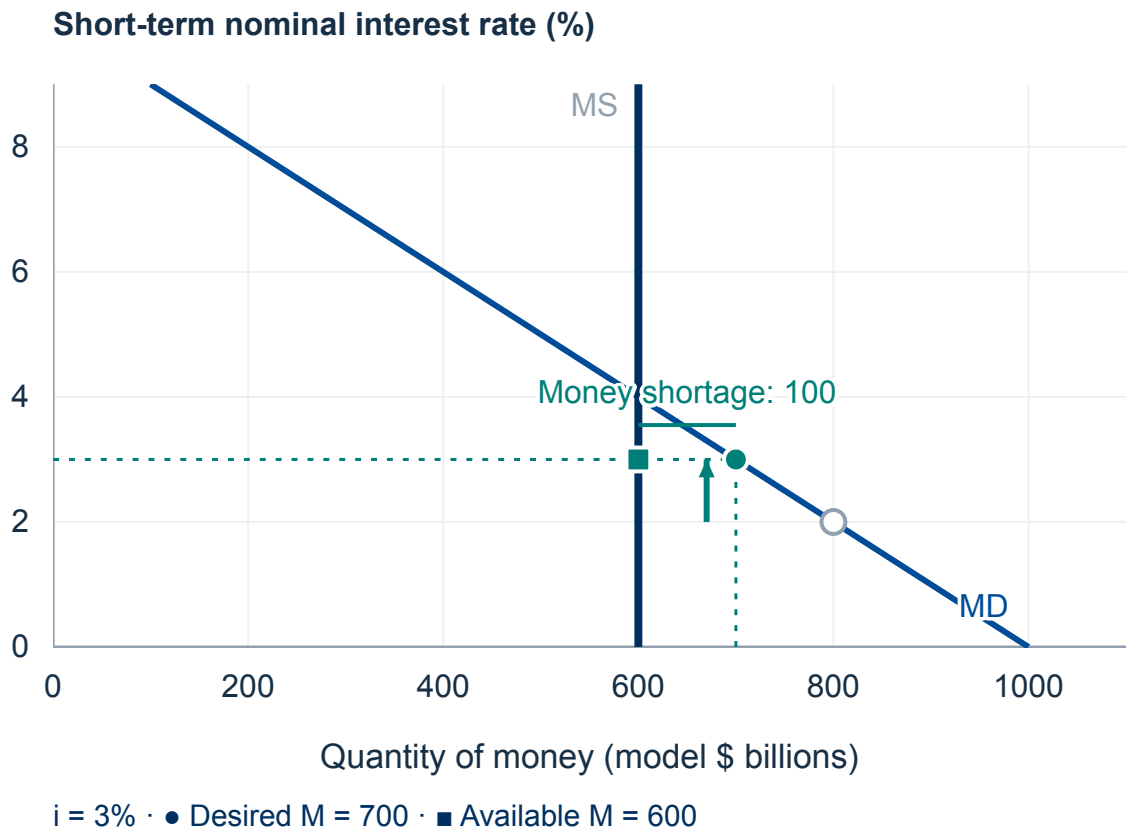
A rate below equilibrium

Suppose i = 2%: MD = 800 but MS = 600. People want more liquid balances.

A shortage of money balances

Households sell bonds to build balances for rent and bills; not everyone can raise money holdings at once.

Adjustment from a Low Interest Rate



i = 3% · ● Desired M = 700 · ■ Available M = 600

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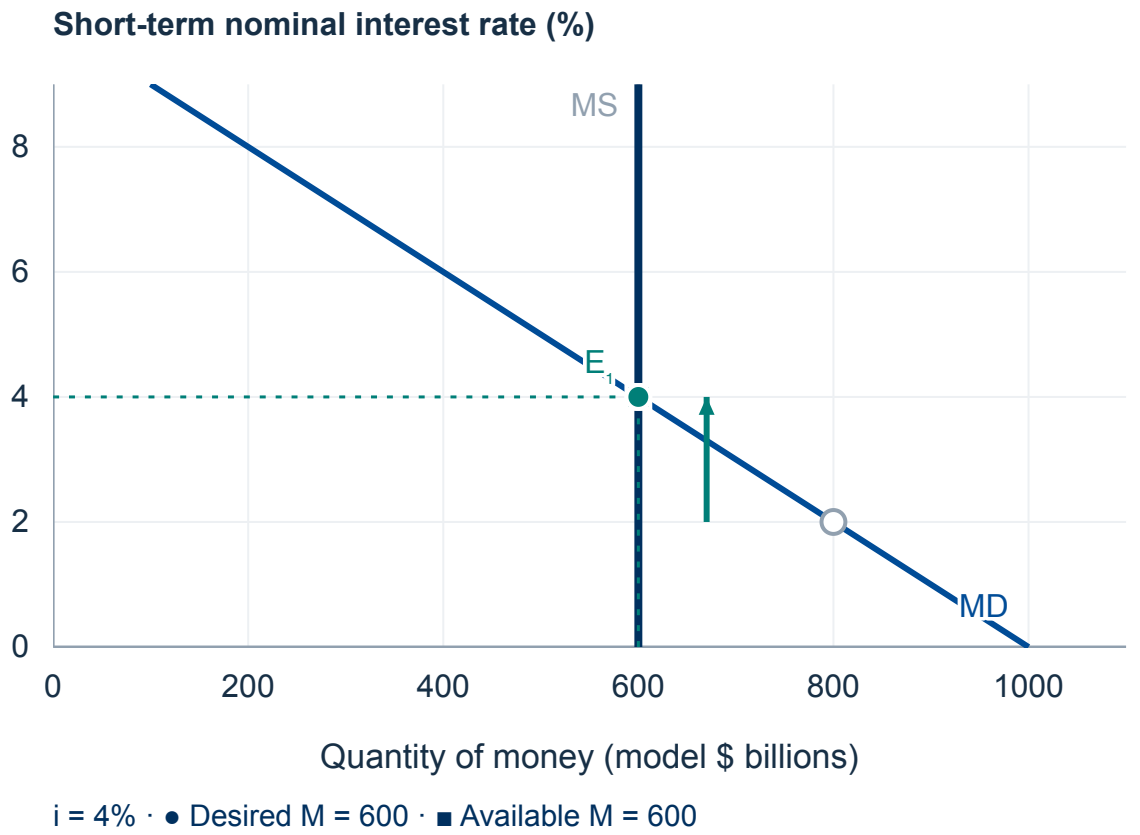
A shortage of money balances

Households sell bonds to build balances for rent and bills; not everyone can raise money holdings at once.

Bond prices fall; yields rise

Bond prices fall and yields rise. A $100 one-year payoff falls from $98.04 to $96.15.

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Bond prices fall and yields rise. A $100 one-year payoff falls from $98.04 to $96.15.

Money-market equilibrium

At i = 4%, desired and available money both equal 600.

Money-Market Effects

Policy in the model	Money supply	Bond price	Short-term rate
Expansionary purchase	Increases	Rises	Falls
Contractionary sale	Decreases	Falls	Rises

Hold money demand, the price level, real GDP, and money’s own return fixed.

Money-Market Checkpoint

1. Money supply increases while money demand is unchanged. Which curve moves, and what happens to the equilibrium short-term rate?

2. At a rate below equilibrium, desired money exceeds the available stock. What adjustment removes the shortage?

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> MS moves right. At the old rate there are excess money balances; asset purchases raise bond prices and lower the equilibrium rate.

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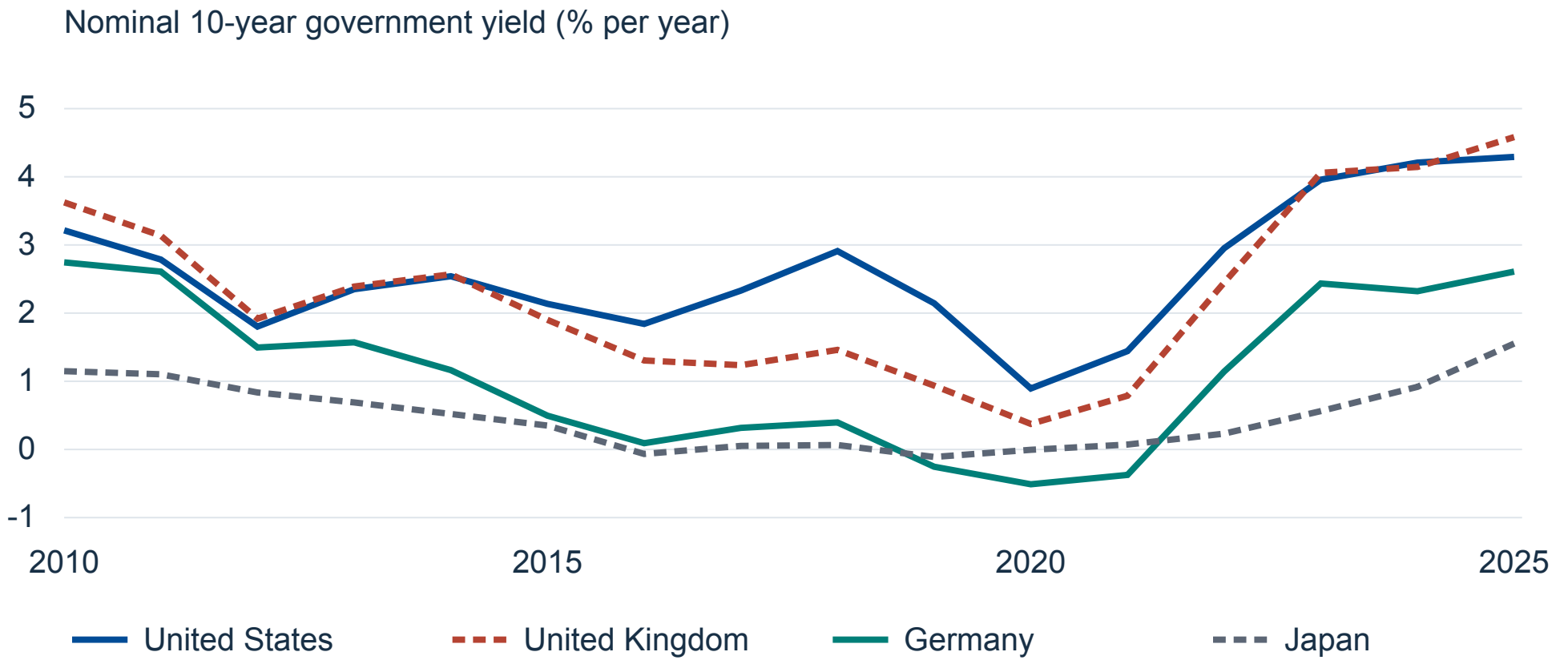
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2. At a rate below equilibrium, desired money exceeds the available stock. What adjustment removes the shortage?

> People sell bonds to obtain money. Bond prices fall, yields rise and the quantity of money demanded falls toward the fixed stock.

Long-Term Interest Rates across Countries



Annual averages, 2010–2025; own-currency yields. National benchmark definitions differ.

OECD (2026), Financial market statistics: long-term interest rates (accessed November 2, 2026) · OECD data terms, section 3

Comparing One-Year and Two-Year Investments

Invest $600 for two years	Year 1 rate	Year 2 rate
Two-year contract, annual compounding	5% fixed	5% fixed
Roll one-year investments: P5	4%	2% scenario
Roll one-year investments: P6	4%	8% scenario

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Maturity and Interest Rates

- A longer-term yield reflects the expected path of short-term rates and compensation for holding longer-term risk.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

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Maturity and Interest Rates

- A longer-term yield reflects the expected path of short-term rates and compensation for holding longer-term risk.
- A change in today’s policy rate may already be anticipated; a 10-year yield need not change by the same amount.
- Longer maturity is different from the macroeconomic long run. A two-year investment is longer than a one-year investment; the data chart uses a 10-year benchmark.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

15.3 Monetary Policy and Aggregate Demand

- Explain how monetary policy affects consumption, investment, and aggregate demand.
- Interpret policy-induced AD shifts while holding short-run aggregate supply fixed.
- Demonstrate expansionary and contractionary effects on output and the price level.

Fiscal and Monetary Policy

Policy choice	Main route	When especially useful
Monetary policy	Changes financial conditions; affects C and I	Broad demand management with room to adjust rates
Fiscal policy	Government purchases, taxes, and transfers	Direct purchases or support targeted to affected groups
A coordinated response	Combines instruments	Severe downturns, especially near the effective lower bound

- Automatic stabilizers can act without a new law; discretionary fiscal measures may take time to enact.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

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- The choice also depends on inflation, implementation delays, debt capacity, and the type of shock. A supply loss is not repaired simply by adding demand.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

From Interest Rates to Aggregate Demand

1 A lower policy rate can ease broader borrowing and financial conditions.

Transmission is conditional: expected inflation, credit access, risk, and confidence also matter.

Federal Reserve Board: policy goals and transmission

From Interest Rates to Aggregate Demand

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Federal Reserve Board: policy goals and transmission

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2. Lower relevant real borrowing costs can encourage consumption and investment.
3. A $10,000 project paying $10,500 next year loses money with a $10,600 repayment but gains with a $10,400 repayment.

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Federal Reserve Board: policy goals and transmission

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1. A lower policy rate can ease broader borrowing and financial conditions.
2. Lower relevant real borrowing costs can encourage consumption and investment.
3. A $10,000 project paying $10,500 next year loses money with a $10,600 repayment but gains with a $10,400 repayment.
4. At a given price level, higher C and I raise planned expenditure: AD shifts right.

Transmission is conditional: expected inflation, credit access, risk, and confidence also matter.

Federal Reserve Board: policy goals and transmission

Aggregate Demand Checkpoint

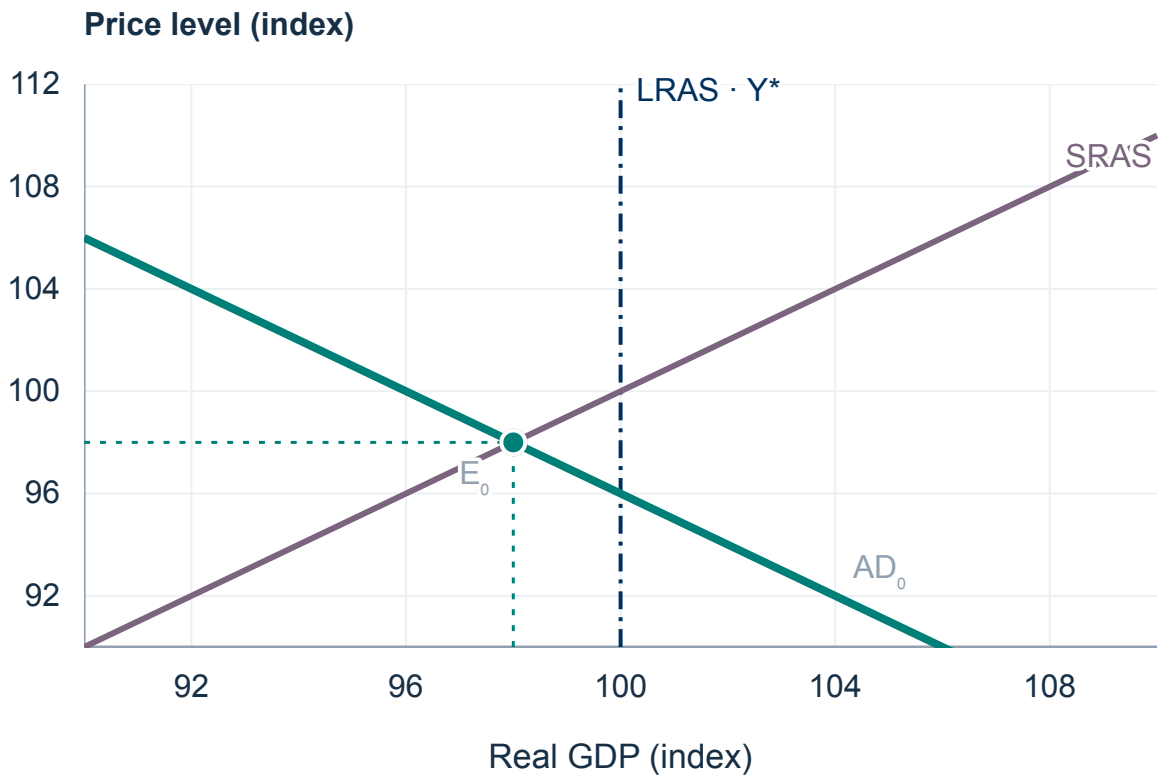
The economy is below potential. Monetary easing lowers relevant real borrowing costs, and SRAS is fixed. Predict the direction of C, I, AD, output and the price level.

Aggregate Demand Checkpoint

The economy is below potential. Monetary easing lowers relevant real borrowing costs, and SRAS is fixed. Predict the direction of C, I, AD, output and the price level.

> Interest-sensitive C and I rise; AD shifts right. Short-run output and the price level rise. The next two graphs show the expansionary case and its contractionary counterpart.

Expansionary Policy and Aggregate Demand



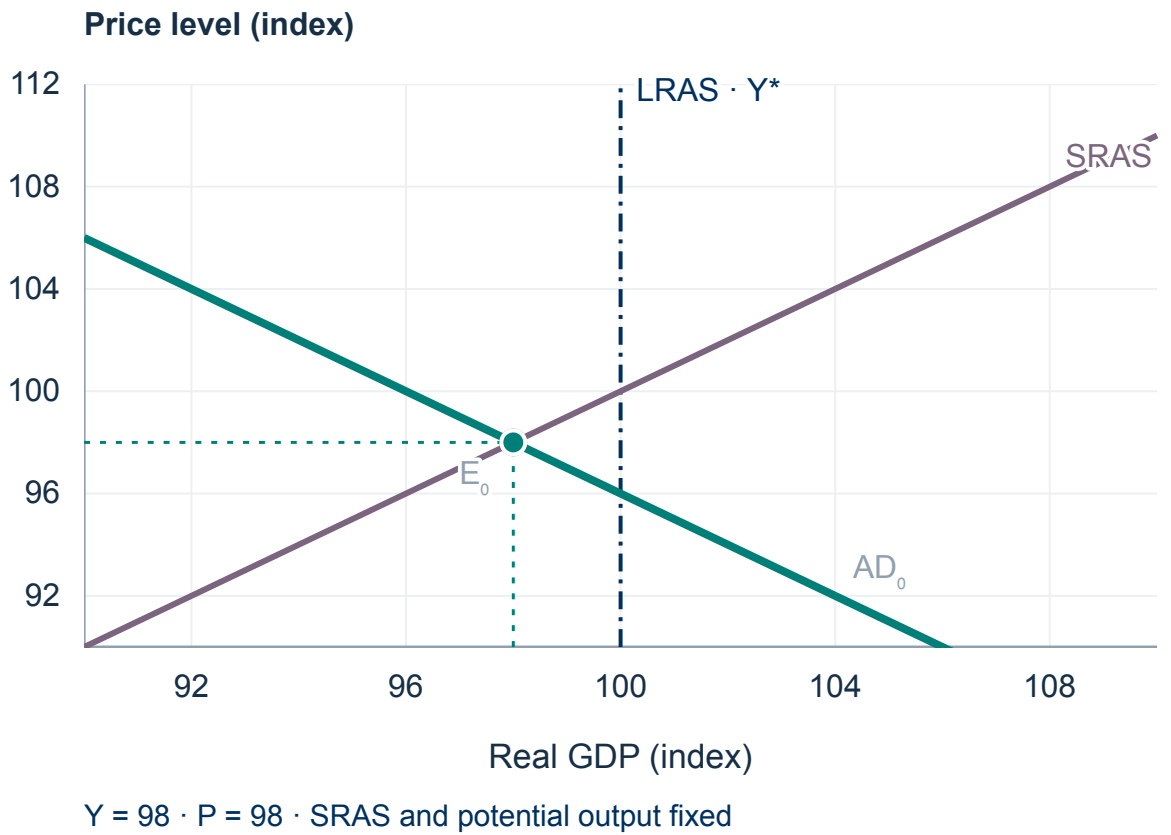
Y = 98 · P = 98 · SRAS and potential output fixed

Original index model: AD shifts; SRAS and LRAS stay fixed.

A negative output gap

Begin below potential output: $Y_0 = 98$, while $Y^* = 100$.

Expansionary Policy and Aggregate Demand



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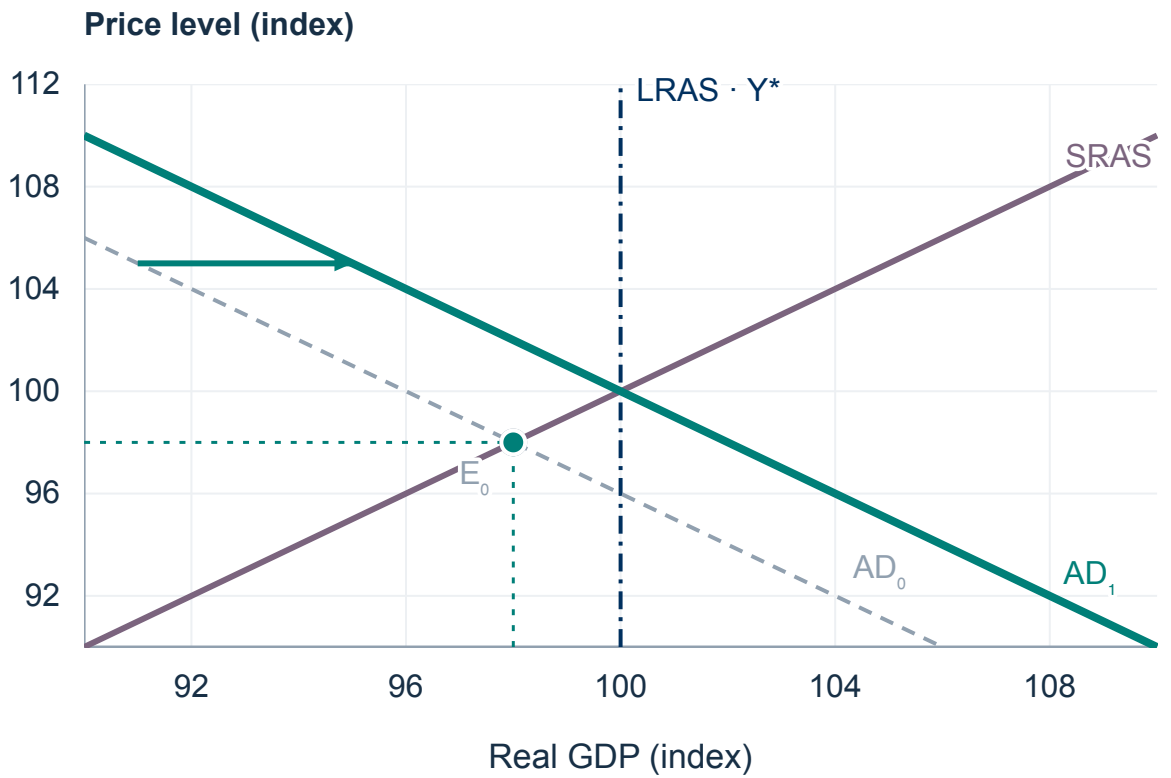
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Financial conditions ease

Easier financial conditions encourage interest-sensitive C and I.

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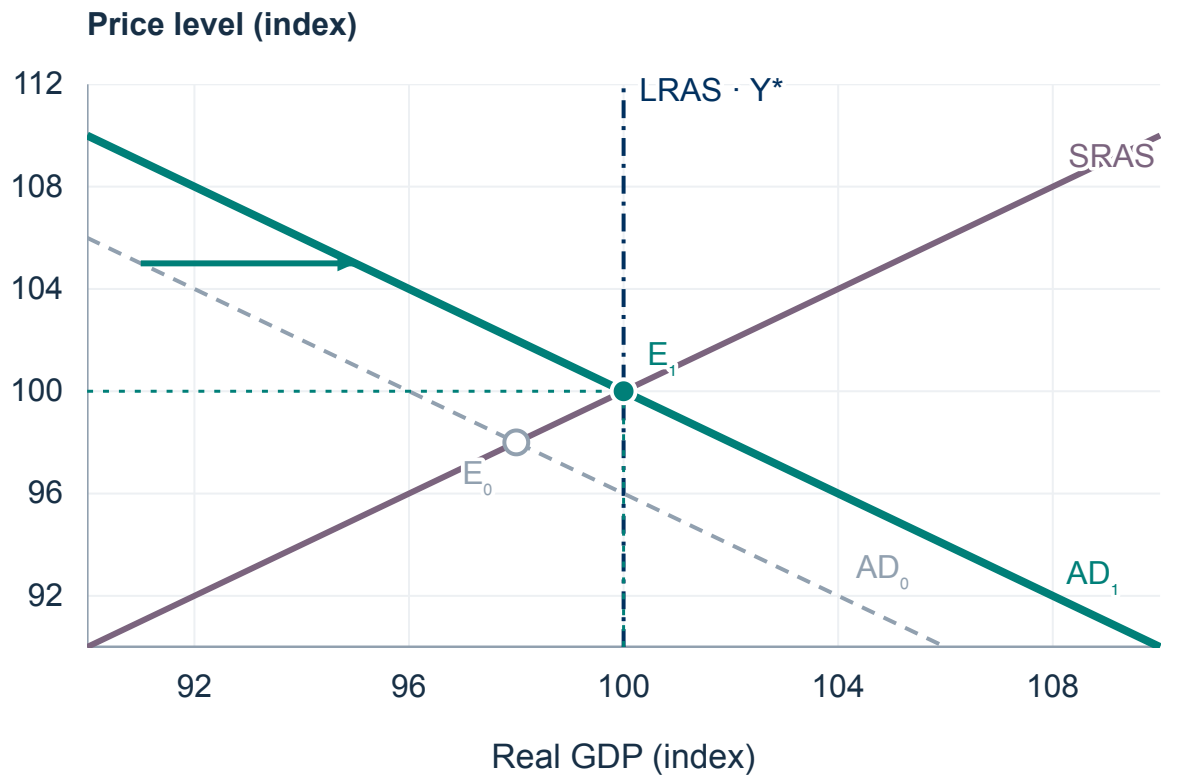
Financial conditions ease

Easier financial conditions encourage interest-sensitive C and I.

Aggregate demand moves right

AD shifts right, from AD_0 to AD_1; SRAS and LRAS stay fixed.

Expansionary Policy and Aggregate Demand



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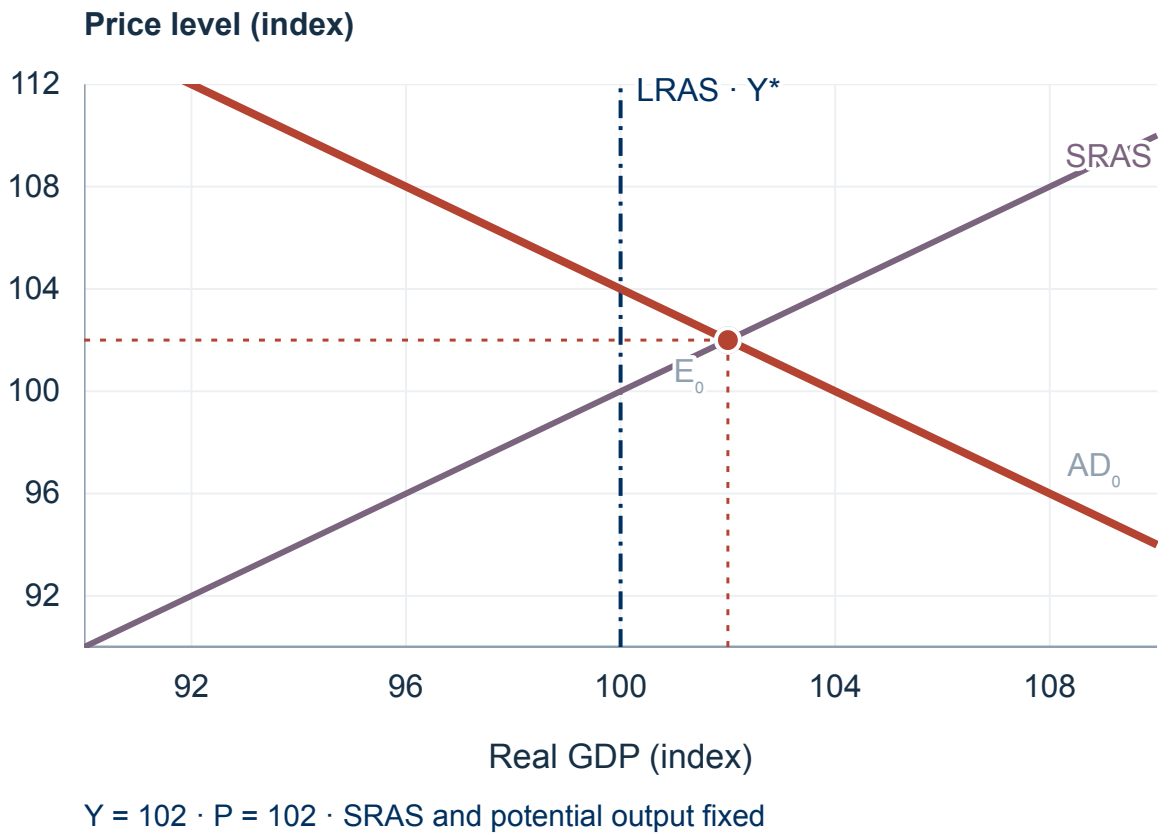
Aggregate demand moves right

AD shifts right, from AD_0 to AD_1; SRAS and LRAS stay fixed.

The short-run equilibrium

In this short-run model, output and the price level rise to 100.

Contractionary Policy and Aggregate Demand



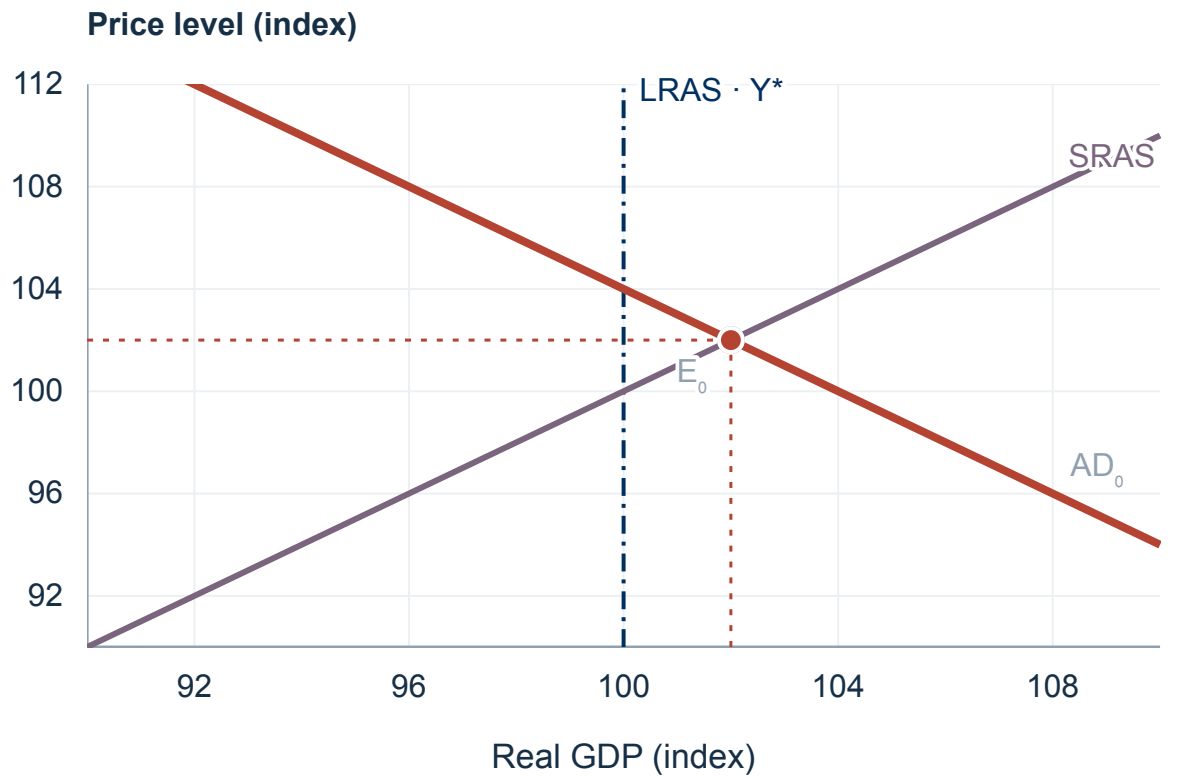
Y = 102 · P = 102 · SRAS and potential output fixed

Original index model: AD shifts; SRAS and LRAS stay fixed.

A positive output gap

Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.

Contractionary Policy and Aggregate Demand



Y = 102 · P = 102 · SRAS and potential output fixed

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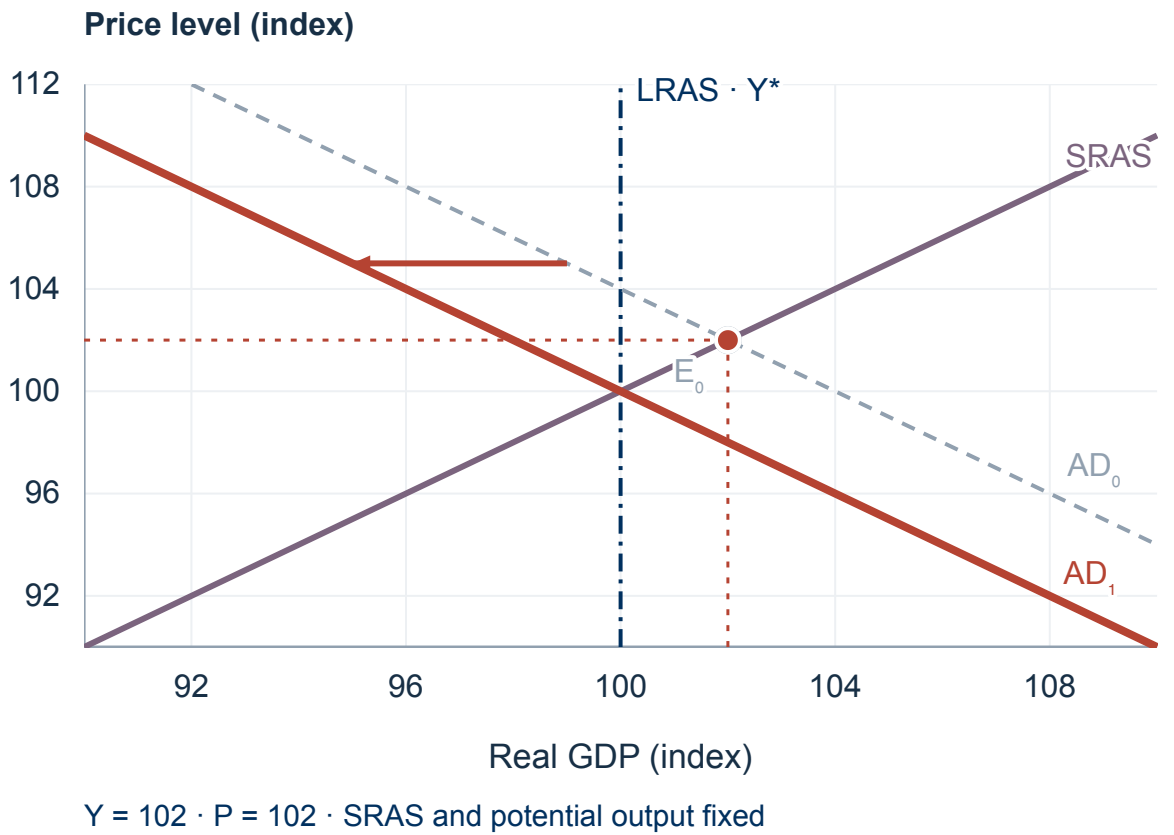
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Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.

Financial conditions tighten

Tighter financial conditions restrain interest-sensitive C and I.

Contractionary Policy and Aggregate Demand



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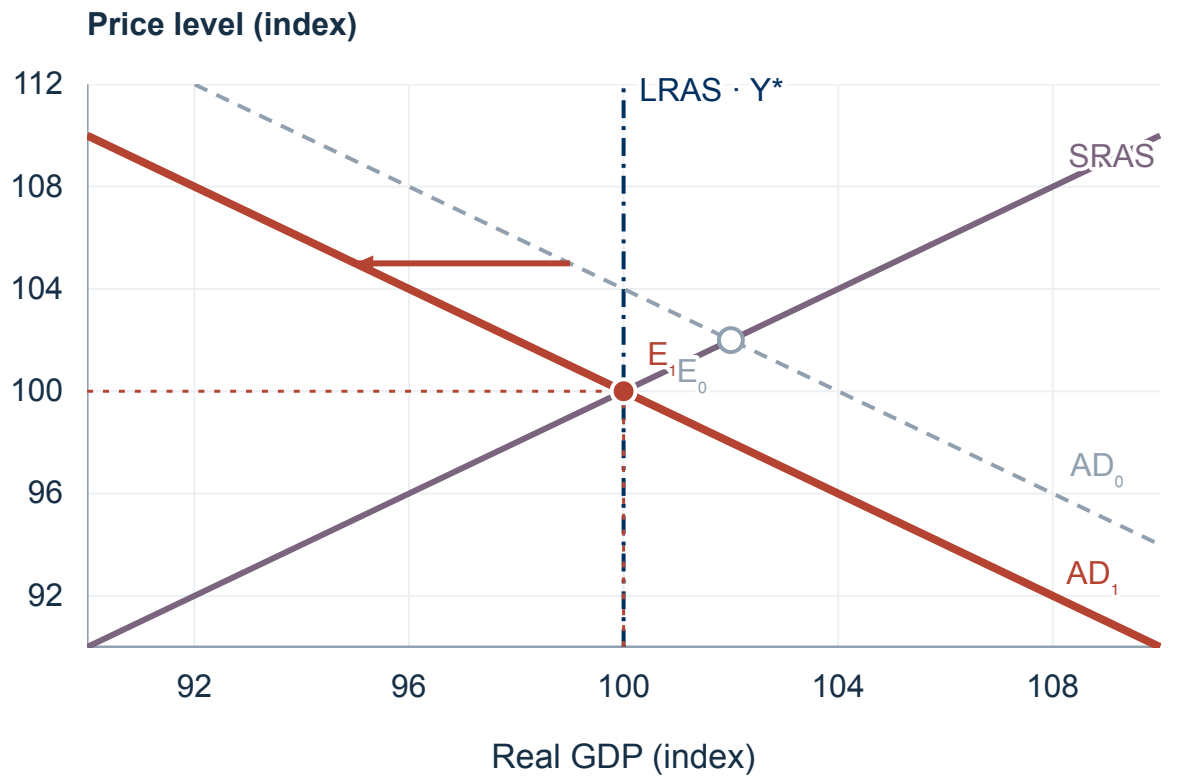
Financial conditions tighten

Tighter financial conditions restrain interest-sensitive C and I.

Aggregate demand moves left

AD shifts left, from AD_0 to AD_1; SRAS and LRAS stay fixed.

Contractionary Policy and Aggregate Demand



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Financial conditions tighten

Tighter financial conditions restrain interest-sensitive C and I.

Aggregate demand moves left

AD shifts left, from AD_0 to AD_1; SRAS and LRAS stay fixed.

The short-run equilibrium

In this short-run model, output and the price level are lower than without the tightening.

Monetary Policy: Short-Run Effects

Channel	Expansionary policy	Contractionary policy
Money-market model	MS right; short-term rate falls	MS left; short-term rate rises
Financial conditions	Relevant borrowing costs can fall	Relevant borrowing costs can rise
Aggregate demand	C and I rise; AD shifts right	C and I fall; AD shifts left
With SRAS fixed	Output and price level rise	Output and price level fall relative to the alternative

Effects depend on the shock and financial transmission. Potential output stays fixed in these short-run comparisons.